

Putnam and Beyond

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Problems taken from the book by the same name as the title, by Andreescu and Gelca.

Chapter 2.2 — Problem 147

Let $P(x)$ be a polynomial of odd degree with real coefficients. Show that $P(P(x)) = 0$ has at least as many real roots as the equation $P(x) = 0$, counted without multiplicities.

Solution

It is well-known that any polynomial with odd degree has at least one real root. Suppose $r_1 < r_2 < \dots < r_k$ are the real roots of $P(x)$. Consider the polynomials $P_i(x) = P(x) - r_i$, ($i = 1, 2, \dots, k$). Noticing that $\deg P_i = \deg P$ we conclude that each $P_i(x)$ has at least one real root. Suppose that $\alpha_i \in \mathbb{R}$ is one of the real roots of P_i . Then

$$P(\alpha_i) = r_i$$

$$P(P(\alpha_i)) = P(r_i) = 0.$$

So $\alpha_1, \alpha_2, \dots, \alpha_k$ are roots of $P(P(x))$. Hence $P(P(x))$ has at least k real roots (counted without multiplicity).

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Chapter 2.2 — Problem 151

Let $P(x)$ be a polynomial of degree n . Knowing that

$$P(k) = \frac{k}{k+1}, \quad k = 0, 1, \dots, n$$

find $P(m)$ for $m > n$.

Solution

Define $g(x) = (x+1)P(x) - x$. This is a polynomial of degree $(n+1)$ with roots $0, 1, \dots, n$. Thus for some $\lambda \in \mathbb{C}$,

$$g(x) = \lambda x(x-1)(x-2) \cdots (x-n).$$

Now we can re-arrange to find that

$$P(x) = \frac{g(x) + x}{x+1}.$$

Now since P is a polynomial of degree $(n+1)$, $(x+1)$ should divide $g(x) + x$. Factor theorem tells us that $g(-1) - 1 = 0$. Now we can compute λ ,

$$\begin{aligned} g(-1) &= \lambda(-1)(-2) \cdots (-n-1) \\ &= \lambda(-1)^{n+1}(n+1)! \end{aligned}$$

Since $g(-1) = 1$ we find that $\lambda = \frac{1}{(-1)^{n+1}(n+1)!}$. So we find that

$$P(x) = \frac{\frac{1}{(-1)^{n+1}(n+1)!} x(x-1) \cdots (x-n) + x}{x+1}$$

or using generalised binomial coefficients

$$P(x) = \frac{1}{x+1} \left(\binom{-x}{n+1} + x \right). \blacksquare$$

Chapter 2.2 — Problem 148

Determine all polynomials $P(x)$ with real coefficients for which there exists a positive integer n such that for all x ,

$$P\left(x + \frac{1}{n}\right) + P\left(x - \frac{1}{n}\right) = 2P(x).$$

Solution

The answer is that all linear polynomials $P(x) = ax + b$ work, $a, b \in \mathbb{R}$. We verify

$$\left[a\left(x + \frac{1}{n}\right) + b\right] + \left[a\left(x - \frac{1}{n}\right) + b\right] = 2(ax + b).$$

Now let's show that polynomials of degree ≥ 2 don't work.

Claim 1: Consider some $p(x) \in \mathbb{R}[x]$, with degree n . Fix $d > 0$. Then $\deg[p(x+d) - p(x)] = n-1$.

Proof: Let $p(x) = a_n x^n + a_{n-1} x^{n-1} + \cdots + a_0$, where $a_n \neq 0$. Note that

$$\begin{aligned} p(x+d) - p(x) &= \sum_{r=0}^n a_r [(x+d)^r - x^r] \\ &= \sum_{r=0}^n a_r \sum_{k=0}^{r-1} \binom{r}{k} x^k d^{r-k} \end{aligned}$$

and so $\deg[p(x+d) - p(x)] = n-1$. $\square$

Suppose that $\deg P = N$. We may re-arrange the given equation to get

$$P\left(x + \frac{1}{n}\right) - P(x) = P(x) - P\left(x - \frac{1}{n}\right)$$

Define $\Delta_n(x) = P\left(x + \frac{1}{n}\right) - P(x)$. Observe $\deg \Delta_n = N-1$. We have

$$\Delta_n(x) - \Delta_{n-1}(x) = 0$$

Noting that $\deg[\Delta_n(x) - \Delta_{n-1}(x)] = N-2 = 0$, we find that $N = 2$. So now suppose that $P(x) = ax^2 + bx + c$. Plugging this into the given equation We have

$$\frac{2a}{n^2} = 0$$

and so $a = 0$. Thus only linear P work. $\blacksquare$